

Md Monirul Islam

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Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
• *B.Sc. in Electrical and Electronic Engineering* March 2018 – May 2023
CGPA: 3.85/4.00 (**3.98/4.00 in the final year**), ranked **25 of 189**, awarded with Honours
Relevant Courses: Electrical & Electronic Circuits, Calculus I & II, Ordinary & Partial Differential Equations, Linear Algebra, Communication System, Digital Signal Processing, Digital Electronics, Power Electronics, Control Systems, Microprocessors and Embedded Systems, Analog Integrated Circuits and Design, VLSI Circuits and Design

Research Interests

- Robotics • Autonomous Systems • Multi-Robot Coordination • Motion Planning • Path Planning • Real-Time Control • Sensor Fusion • Embedded Systems • Robot Perception • Human-Robot Interaction • Field Deployment of Robotic Platforms

Research Contributions & Field Deployments

- **Production Multi-AGV Fleet (live garment factory).** Lead engineer end-to-end for a 3-robot AGV fleet operating continuously alongside hundreds of human workers with no physical separation. Designed FleetCore (ROS 2 dispatcher with per-robot FSMs, mutual exclusion at shared resources, layered timeouts, fail-closed substrate), a custom 5-state EKF with camera-latency compensation for AprilTag-based localization, and a fully decentralized A* path-planner with peer-aware timestamped reservations (three conflict checks: node, edge-swap, final-cell). Months of zero-collision production operation. *Productised as Cybernetics CyberFleet.*
- **7-DOF Mobile Manipulator (ROS 2 / MoveIt 2).** Owned the full control / software stack. Replaced MoveIt's KDL solver with a custom closed-form analytical IK; added smooth turret scaling for the home-pose singularity and an FK-anchored goal accumulator that cut un-commanded cross-axis motion $5\times$. 3,500 lines of Python across coordinator, IK, FK, SBus decoder, and a 40-test unit suite; full URDF/SRDF + ros2_control integration. *Cybernetics "Jontro Soinik 2.0".*
- **Indigenous EOD ROV Family (Jontro Soinik v1 & v2).** v1 (Mali UN deployment, 2024) and v2 (Congo UN deployment, 2025). Designed v1's star-topology communication network and its S-curve arm-motion profile; for v2, designed the custom multi-drop bus and the differential-drive speed-synchronization layer (subject of a first-author manuscript). All controller PCBs designed in Altium / KiCad and milled in-house on a 3-axis CNC. *Cybernetics.*
- **Heavy-Lift Multirotor (50 kg payload).** Designed BMS, propulsion stack, frame, and power-distribution PCBs for a heavy-lift cargo / emergency-response drone. The battery + motor sizing study has been ported into a browser-side Drone Battery Optimizer (Pareto search over $Y_s \times Z_p$ pack configurations) published on the portfolio site. *Cybernetics.*
- **Vision-Guided Target-Tracking Drone (SD15-N).** Designed the autonomous target-tracking pipeline: YOLOv11n on a Raspberry Pi 5 closed-loop with a Pixhawk 2.4.8 over MAVLink. *Cybernetics.*
- **Industrial CoreXY 3D Printer (Titan 550).** Two-iteration product build: $510 \times 510 \times 550$ mm volume, ± 0.02 mm repeatability, BigTreeTech + TMC2209, Marlin 2.0; in-house workhorse for every other platform's fixtures. *Cybernetics.*
- **α -Graphyne Ballistic Nanoribbon FET.** BUET undergraduate research with Dr. Mahbub Alam — modelled ballistic quantum transport in α -graphyne nanoribbons; published in *Micromachines* (MDPI), 2023.

Publications

- **Md Monirul Islam** et al., "Peer-Aware Distributed Path Planning for a Production-Deployed Multi-AGV Fleet in a Human-Occupied Garment Factory", *In Preparation*, 2026 (first / corresponding author).
- **Md Monirul Islam** et al., "Experimental Speed Synchronization of a Dual-BLDC Tracked Differential-Drive Vehicle Using a GA-Tuned Bilateral Cross-Coupled Controller on a Dual-MCU Architecture", *Under Review*, 2026 (first / corresponding author).
- **Md Monirul Islam** et al., "A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control", *Under Review*, 2026 (co-author).
- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard", 2024, 27th International Conference on Computer and Information Technology (ICCIT), IEEE. doi: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)
- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, "Performance Analysis of an α -Graphyne Nano-Field Effect Transistor", *Micromachines*, MDPI, 2023. doi: [10.3390/mi14071385](https://doi.org/10.3390/mi14071385)

Professional Experience

- **Research and Development Unit, Cybernetics Hi-Tech Solutions (Pvt) Ltd** Dhaka, Bangladesh
R&D Engineer and Team Lead (June 2023 – present)
 - **Production AGV fleet (live garment factory).** Architected and shipped a 3-robot ROS 2 AGV system: FleetCore dispatcher with per-robot FSMs, FIFO job queue, mutual exclusion on shared resources, layered timeouts, and a fail-closed heartbeat / directory substrate. Per-robot 5-state EKF (pose + velocity biases) with rewind-and-replay camera-latency compensation for AprilTag fixes; fully decentralized A* path planning with peer-aware timestamped reservations and continuous timestamp maintenance.
 - **7-DOF mobile manipulator (ROS 2 / MoveIt 2).** Custom closed-form analytical IK replacing MoveIt's KDL solver, smooth turret-scaling for the home-pose singularity, and an FK-anchored goal accumulator reducing un-commanded cross-axis motion by $\sim 5\times$. 40-test unit suite, full URDF/SRDF + ros2_control integration.
 - **EOD ROV electrical & firmware (Jontro Soinik v1, Mali UN; v2, Congo UN).** v1: star-topology communication; jerk-limited S-curve arm-motion profile across 6 DOFs. v2: custom multi-drop bus replacing the v1 star; differential-drive speed-synchronization layer keeping left / right drives locked through acceleration / cruise / deceleration. All control & comms PCBs designed in Altium / KiCad and milled in-house on a 3-axis CNC — no external fab in the loop.
 - **Heavy-lift multirotor (50 kg payload, 30 min, 15 km).** Custom BMS, propulsion sizing, frame structural design (SolidWorks), and Altium power-distribution PCBs. Published an in-browser "Drone Battery Optimizer" (Pareto search over series-parallel pack configurations) on the portfolio site.
 - **SD15-N target-tracking surveillance drone.** Autonomous-tracking pipeline integrating YOLOv11n on a Raspberry Pi 5 with a Pixhawk 2.4.8 flight controller over MAVLink.
 - **Titan 550 industrial CoreXY 3D printer.** Two-iteration product build ($510 \times 510 \times 550$ mm, ± 0.02 mm repeatability) used in-house for every Cybernetics platform's fixtures and prototypes.
 - **Other production work:** hybrid VTOL quad-plane prototype; industrial automation project for real-time pet-food sorting; operator-training sessions and field hand-over of EOD ROVs to receiving military units.
- **Ulkasemi Pvt. Limited** Dhaka, Bangladesh
Intern (Dec 2022 – Mar 2023)
 - Worked on IC design using TSMC 180 nm and GPDK 90 nm technologies.
 - Developed a Wishbone interface and validated a high-speed BiFET Op-Amp in Verilog.
- **Walton Hi-Tech Industries Limited** Dhaka, Bangladesh
Industrial Trainee (Nov 2022)
 - Observed and analyzed PCB manufacturing and EV assembly workflows.
 - Gained exposure to industrial electronics production workflows.

Selected Undergraduate Projects

- **Real-time traffic speed estimation with YOLOv2:** Trained YOLOv2 in MATLAB to detect vehicles and estimate per-vehicle speed from monocular highway video.
- **6-DOF robotic arm (analytical IK):** Designed in SOLIDWORKS, 3D-printed entirely on a self-built hobby printer using custom 3D-printed planetary gearboxes; closed-form analytical inverse-kinematics control on stepper motors.
- **Autonomous GPS-waypoint delivery drone:** Mechanical design in SOLIDWORKS, Arduino flight controller with PID stabilisation, GPS-waypoint navigation, and a custom payload-deployment mechanism.
- **Self-built hobby 3D printer:** Designed every component in SOLIDWORKS, fabricated the frame on a laser cutter, Arduino Mega controller running custom Marlin firmware. The workhorse that produced every other undergraduate project's parts.
- **RV32I multicycle RISC-V processor + Wishbone memory controller:** Custom Verilog implementation with full testbench verification; physical-layout exploration in Cadence at TSMC 180 nm.
- **OMR machine from discrete digital logic:** Optical-mark recognition built entirely from discrete NAND / RAM / MUX / Comparator / LDR components — no microcontroller in the loop.
- **Regulated boost converter (TSMC 180 nm, Cadence):** Schematic capture and physical layout of a voltage-regulated DC-DC boost converter.
- **Closed-loop induction-motor speed control:** V/f scalar control with SVPWM inverter modulation in MATLAB / Simulink.
- **Particle-concentration light-scattering sensor:** Laser + photodetector light-scattering setup with MATLAB-side calibration curves and signal-conditioning circuit.
- **Real-time environmental monitoring:** Solar-powered IoT node logging temperature, humidity, and CO₂ over GSM; Arduino + Autodesk Eagle PCB.
- **Laboratory variable DC power supply:** Linear-regulator design with current limiting and short-circuit protection; MATLAB efficiency study.
- **Electrical-services design (seven-storey building):** End-to-end electrical design: load estimation, panel scheduling, cable sizing, lighting / power layouts, riser diagram, earthing scheme.

Skills Summary

- **Robotics & Middleware:** ROS 2 (Jazzy), MoveIt 2, ros2.control, JointTrajectoryController, URDF / Xacro / SRDF, AprilTag, MAVLink, dronekit, pymavlink
- **Embedded & MCU:** STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, µVision Keil, BigTreeTech 32-bit boards
- **Control & Algorithms:** Closed-form analytical IK, Extended Kalman Filter, PID (cascaded, GA-tuned), S-curve motion profiling, distributed A* path planning, peer-aware spatiotemporal reservations
- **Flight Control & Vision:** ArduPilot, iNav, Betaflight, Pixhawk, YOLOv11n, OpenCV
- **Simulation & Design Software:** MATLAB / Simulink, Cadence Virtuoso, PSPICE, PSAF, Icarus Verilog, GTKWave
- **PCB & Circuit Design:** Altium Designer, KiCad, Autodesk Eagle, Proteus; in-house 3-axis CNC milling for prototype boards
- **CAD & Mechanical Design:** SolidWorks
- **3D Printing & CNC:** Marlin (customised), Ultimaker Cura, Creality CR-6 SE; FlatCAM, Universal G-code Sender, GRBL
- **Programming Languages:** Python, C, Embedded C, Verilog
- **Documentation:** Microsoft Office Suite, L^AT_EX, Jekyll (al-folio)
- **Soft Skills:** Leadership, Project Management, Field Commissioning, Operator Training, Technical Writing

Honors and Awards

- University Merit Scholarship, BUET (three times — January 2019, January 2020, January 2022)
 - Awarded to students with outstanding academic record (GPA 3.90+) in a semester
- Dean's List Award, BUET (four times — all academic years)
 - Awarded to students obtaining an average GPA of 3.75+ in two consecutive semesters of an academic year
- University Stipend (three times — January 2018, July 2018, July 2021)
- Higher Secondary Certificate (HSC-2017) Board Scholarship
- Runners-up, Bangabandhu Creative Talent Hunt 2014, Pabna Division

Extra-curricular Activities

- Affiliate Member, BUET Robotics Society (BRS). (Jan 2022 – Mar 2023)
- Affiliate Member, Team Interplanetary – BUET Mars Rover Team. (Jun 2022 – Mar 2023)

Field Deployments

- 3-robot AGV fleet – live production at The Urmi Group RMG facility, Dhaka, 2025 – present
- Jontro Soinik v1 EOD ROV – UN Peacekeeping Mission, Mali, 2024 (Bangladesh Army → Peruvian Armed Forces handover)
- Jontro Soinik v2 EOD ROV – tested by Bangladesh Army, set for UN Peacekeeping deployment in the Republic of Congo, 2025

References

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